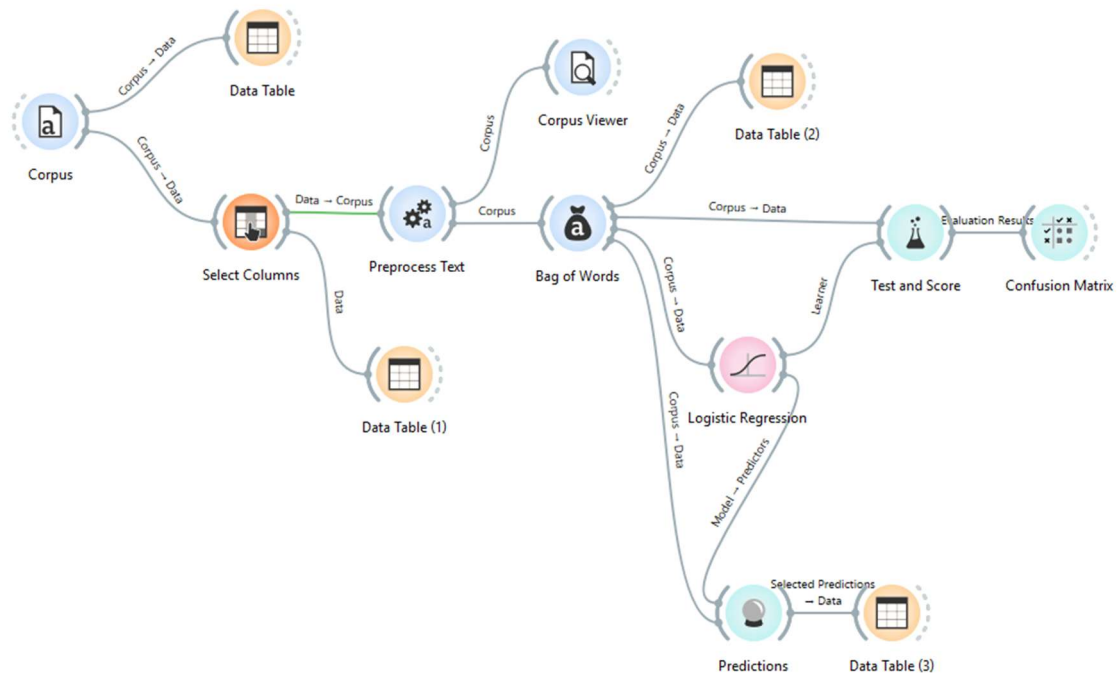


Módulo 6 — Classificação de Texto com Regressão Logística

Prática no Orange.

Use ***Election Tweets 2016***

Abaixo a versão final do pipeline:



1. Carregar dados

Abrir o Orange 3.40

Inserir o widget **Corpus** (precisa do “Options > Add-on > Text 1.16.3”)

Selecionar o corpus ***election-tweets-2016.tab***

Inserir o widget **Data Table**

Conectar **Corpus → Data Table**

Abrir o **Data Table** para visualizar os dados

ATENÇÃO - Caso os dados exibidos não correspondam ao dataset selecionado (por exemplo, apareçam colunas de um laboratório anterior), pode haver persistência de estado no Orange. Nesse caso, acesse **Options → Reset Widget Settings...** e recarregue o widget. Se necessário, recrie o fluxo (**File → Data Table**) para garantir que os dados sejam atualizados corretamente.

Perguntas:

Qual é a variável alvo?

Quantos registros existem no corpus?

Qual coluna contém o texto principal?

2. Selecionar colunas

Inserir o widget **Select Columns**

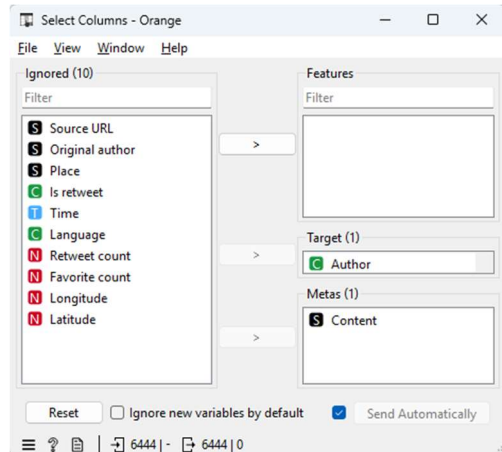
Conectar **Corpus** → **Select Columns**

Abrir o **Select Columns**

Mover **Author** para **Target**

Mover **Content** para **Metas**

Mover todas as demais variáveis para **Ignored**



Inserir o widget **Data Table**

Conectar **Select Columns** → **Data Table**

Abrir o **Data Table** para validar a seleção (apenas duas colunas restam...)

Perguntas:

Quais variáveis permanecem após a seleção?

Qual o tipo do problema após a definição do target?

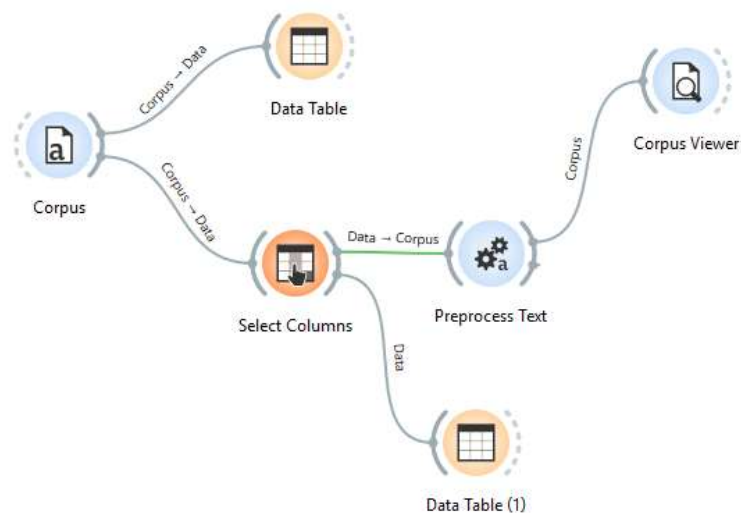
3. Pré-processar o texto

Inserir o widget **Preprocess Text**

Conectar **Select Columns** → **Preprocess Text**

Inserir o widget **Corpus Viewer**

Conectar **Preprocess Text** → **Corpus Viewer**



A ideia agora é aplicar algumas transformações.

Abrir o widget **Preprocess Text**

Adicionar **Transformation**

Marcar **Lowercase** e marcar **Remove URLs**

Adicionar **Tokenization**

Selecionar **Regexp** e informar `\w+` (vai ignorar espaços, pontuações e símbolos)

Adicionar **Normalization**

Selecionar **WordNet Lemmatizer**

Adicionar **Filtering**

Marcar **Stopwords** e selecionar **English**

Abrir o **Corpus Viewer** para visualizar o texto pré-processado

The image shows the configuration window for the 'Preprocess Text' widget, divided into four sections: Transformation, Tokenization, Normalization, and Filtering.

- Transformation:** Includes checkboxes for 'Lowercase' (checked), 'Remove accents', 'Parse html', and 'Remove urls' (checked).
- Tokenization:** Includes radio buttons for 'Word Punctuation', 'Whitespace', 'Sentence', 'Regexp' (selected), and 'Tweet'. A text field next to 'Regexp' contains the pattern '\w+'.
- Normalization:** Includes radio buttons for 'Porter Stemmer', 'Snowball Stemmer', 'WordNet Lemmatizer' (selected), 'UDPipe Lemmatizer', and 'Lemmagin Lemmatizer'. Each has a 'Language' dropdown menu set to 'English'. There is also an unchecked checkbox for 'UDPipe tokenizer'.
- Filtering:** Includes checkboxes for 'Stopwords' (checked), 'Lexicon', 'Numbers', 'Regexp', 'Document frequency', 'Most frequent tokens', and 'POS tags'.
 - 'Stopwords' has a dropdown set to 'English' and a '(none)' dropdown.
 - 'Lexicon' has a '(none)' dropdown.
 - 'Numbers' has an unchecked checkbox for 'Includes Numbers'.
 - 'Regexp' has a text field with a complex pattern.
 - 'Document frequency' has radio buttons for 'Relative' (selected) and 'Absolute'. 'Relative' has two numeric input fields set to 0,10 and 0,90. 'Absolute' has two numeric input fields set to 1 and 10.
 - 'Most frequent tokens' has a numeric input field set to 100.
 - 'POS tags' has a text field containing 'NOUN,VERB'.

Perguntas:

No processo de filtragem de texto, qual é a diferença entre utilizar Stopwords, Lexicon, Regexp, Document Frequency, Most Frequent Tokens e POS Tags?

A ordem das etapas no pré-processamento influencia o resultado final?

Qual a diferença entre definir essa ordem em uma ferramenta visual como o Orange e em um pipeline explícito (ex: notebook em Python)?

4. Vetorização do texto

- Inserir o widget **Bag of Words**
- Conectar **Preprocess Text** → **Bag of Words**
- Inserir o widget **Data Table**
- Conectar **Bag of Words** → **Data Table**
- Abrir o **Data Table** para visualizar a matriz numérica

File Edit View Window Help			
Info			
6444 instances (no missing data)			
8404 features (sparse, density 100.00 %)			
Target with 2 values			
1 meta attribute			
Variables			
☐ Show variable labels (if present)			
☐ Visualize numeric values			
☐ Color by instance classes			
Selection			
Clear Selection			
☒ Select full rows			
	Author	Content	[...]
1	HillaryClinton	The question in this election: Who can put t...	action=1, better=1, election=1, life=1, make=1, plan=1, put=1, question=1
2	HillaryClinton	Last night, Donald Trump said not paying ta...	call=1, donald=1, know=1, last=1, night=1, paying=1, said=1, smart=1, tax=1, trump=1, unpatriotic=1, was=1
3	HillaryClinton	Couldn't be more proud of @HillaryClinton. ...	command=1, debate=1, hillaryclinton=1, last=1, next=1, night=1, potus=1, proud=1, ready=1, showed=1, visions=1
4	HillaryClinton	If we stand together, there's nothing we can ...	make=1, nothings=1, ready=1, stand=1, sure=1, together=1, votes=1
5	HillaryClinton	Both candidates were asked about how they ...	answer=1, asked=1, candidate=1, confront=1, injustice=1, one=1, racial=1, real=1
6	realDonaldTrump	Join me for a 3pm rally - tomorrow at the Mi...	3pm=1, america=1, bluffs=1, center=1, council=1, iowa=1, join=1, mid=1, rally=1, ticket=1, tomorrow=1
7	HillaryClinton	This election is too important to sit out. Go t...	election=1, go=1, h=1, important=1, make=1, nationalvoterregistrationday=1, registered=1, sit=1, sure=1
8	HillaryClinton	When Donald Trump goes low...register to v...	donald=1, go=1, low=1, register=1, trump=1, votes=1
9	realDonaldTrump	Once again, we will have a government of, b...	government=1, join=1, movement=1, people=1, today=1
10	HillaryClinton	3) Has Trump offered a single proposal to re...	3=1, business=1, friction=1, has=2, hillaryclinton=1, offered=1, proposal=1, reduce=1, single=1, starting=1, trump=1
11	HillaryClinton	The election is just weeks away. Check if you...	away=1, check=1, cls=1, election=1, registered=1, take=1, vote=1, weeks=1
12	realDonaldTrump	On National #VoterRegistrationDay, make su...	make=1, makeamericagreatagain=1, national=1, registered=1, sure=1, votes=1, voterregistrationday=1
13	realDonaldTrump	Hillary Clinton's Campaign Continues To Ma...	campaign=1, claims=1, clinton=1, continues=1, disclosures=1, sales=1, foundations=1, hillary=1, makes=1
14	realDonaldTrump	CNBC, Time magazine online polls say Dona...	cnbcc=1, debate=1, donald=1, first=1, maga=1, magazine=1, online=1, polls=1, presidential=1, say=1, times=1, trump=1, via=1, washtimes=1
15	HillaryClinton	Donald Trump lied to the American people a...	58=1, americans=1, counted=1, debate=1, donald=1, first=1, least=1, lied=1, people=1, presidential=1, times=1, trump=1
16	realDonaldTrump	Great afternoon in Little Havana with Hispan...	afternoon=1, community=1, great=1, havana=1, hispanics=1, imwithyou=1, leader=1, little=1, support=1, thanks=1
17	realDonaldTrump	In the last 24 hrs. we have raised over \$13M f...	13m=1, 24=1, america=1, call=1, day=1, donation=1, going=1, hrs=1, last=1, maga=1, national=1, online=1, raised=1, still=1, thanks=1
18	HillaryClinton	"She gained about 55 pounds in...9 months..."	55=1, 9=1, eating=1, gained=1, like=1, machine=1, man=1, month=1, pounds=1, president=1, trump=1, wa=1, wants=1
19	HillaryClinton	It's #NationalVoterRegistrationDay. Celebrat...	celebrates=1, nationalvoterregistrationday=1, registering=1, votes=1
20	HillaryClinton	"I love this country. I'm proud of this countr...	brings=1, country=2, hillary=1, leader=1, love=1, lovetrumpshates=1, people=1, proud=1, together=1, wants=1
21	HillaryClinton	We don't want to turn against each other. W...	another=1, big=1, country=1, goal=1, one=1, set=1, strongertogether=1, turn=1, want=2, works=1
22	HillaryClinton	"What we hear from my opponent is danger...	dangerously=1, hear=1, hillary=1, incoherent=1, matters=1, opponent=1, saying=1, unclear=1, words=1
23	HillaryClinton	One candidate made it clear he wasn't prepa...	b=1, candidate=1, clear=2, debate=1, last=1, made=2, night=1, one=1, prepared=2
24	HillaryClinton	"What kind of a person would want to root f...	9=1, family=1, hillary=1, home=1, kind=1, losing=1, million=1, never=1, one=1, person=1, president=1, roots=1, trump=1, want=1, would=1
25	HillaryClinton	"I don't think that any family should have to ...	10=1, care=1, child=1, family=1, hillary=1, income=1, pay=1, think=1
26	realDonaldTrump	"I'm really glad my dad never had a contract...	contract=1, dad=1, donald=1, glad=1, hillary=1, never=1, really=1, trump=1
27	HillaryClinton	When you work hard, you should not be livi...	hard=1, living=1, poverty=1, work=1
28	realDonaldTrump	Well, now they're saying that I not only won ...	big=1, debate=1, forums=1, last=1, nbc=1, nice=1, night=1, presidential=1, saying=1, well=1
29	HillaryClinton	"We may have a record-setting turnout in th...	biggest=1, could=1, election=1, ever=1, may=1, record=1, setting=1, turnout=2
30	HillaryClinton	"You can go to https://t.co/tTgeqNqYm to ...	go=1, hillary=1, hope=1, make=1, nationalvoterregistrationday=1, registered=1, sure=1

Perguntas:

O que representam as 8404 features geradas?

Isso significa que cada instância (o tweet) possui quantas dimensões?

Por que o número de features é tão alto?

O que significa a matriz ser esparsa?

Qual o impacto de alta dimensionalidade no modelo?

Qual é a entrada (X) e a saída (y) do modelo após o Bag of Words? Ou seja, o que vai para o modelo como entrada?

5. Treinar e avaliar o modelo

Atenção - O Orange executa automaticamente ao conectar os widgets. Se você conecta antes de configurar, ele já roda com os parâmetros padrão. No dia a dia, geralmente (amostragem, proporção, etc.), **depois faça a conexão**.

Inserir o widget Logistic Regression

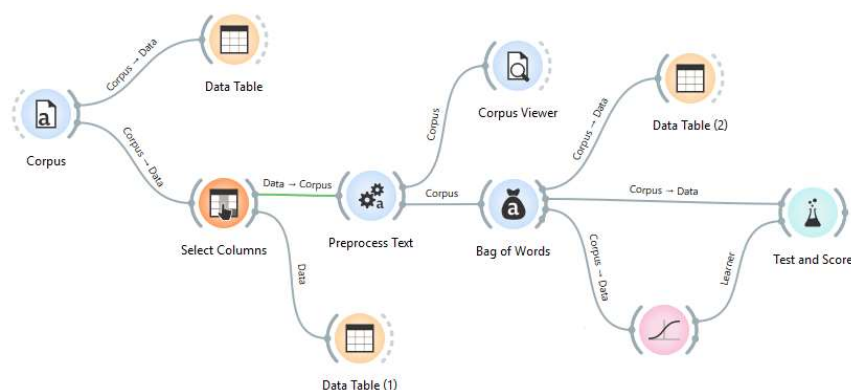
*Conectar **Bag of Words** → **Logistic Regression** (pode dar uma 'travadinha' :)*

Inserir o widget Test and Score

*Conectar **Logistic Regression** → **Test and Score** (vai dar uma 'travadinha' :)*

*Conectar **Bag of Words** → **Test and Score** (vai dar uma 'travadinha' :)*

*Abrir o **Test and Score***



Perguntas:

Qual o objetivo do Test and Score?

O que é estratificação? Ela resolve desbalanceamento?

O que é “Leave-one-out”?

O que é “Test on train data”?

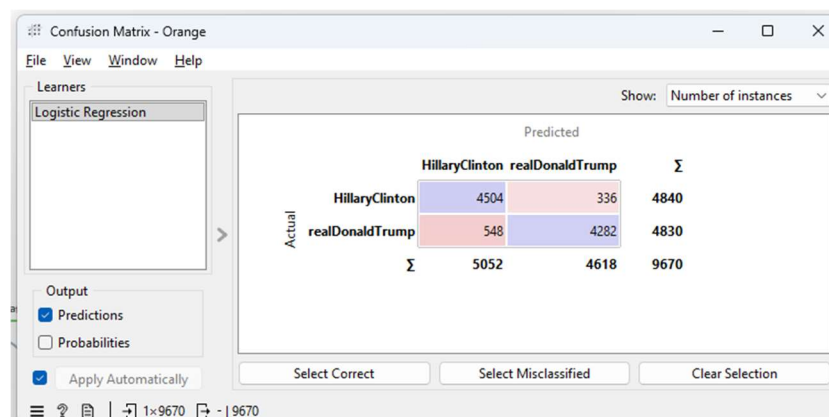
O que é “Test on test data”?

7. Analisar erros

Inserir o widget **Confusion Matrix**

Conectar **Test and Score** → **Confusion Matrix**

Abrir a **Confusion Matrix**



Perguntas:

O que é a matriz de confusão?

O que significa um erro na matriz de confusão?

Na matriz de confusão apresentada, considerando a classe *HillaryClinton* como positiva, quais são os valores de verdadeiro positivo, verdadeiro negativo, falso positivo, falso negativo?

Dada a fórmula da acurácia: $(TP + TN) / (TP + TN + FP + FN)$, qual é o valor da acurácia para a matriz apresentada?

Dada a fórmula da precisão: $TP / (TP + FP)$, qual é o valor da precisão para a matriz apresentada?

8. Analisar previsões

- Inserir o widget *Predictions*
- Conectar *Logistic Regression* → *Predictions*
- Conectar *Bag of Words* → *Predictions*
- Inserir o widget *Data Table*
- Conectar *Predictions* → *Data Table*
- Abrir o *Data Table*

Info	bow-feature	Author	Content	ogistic Regression	egression (Hillar	gression (realDon	ogistic Regression (enc	
6444 instances (no missing data) 8404 features (sparse, density 100.00 %) 5 meta attributes	include skip normalization		True					(...)
Variables	title		True					
☒ Show variable labels (if present)		realDonaldTrump	The ratings at ...	realDonaldTrump	4.10646e-06	0.999996	4.10646e-06	away=1, blow=1, clinton=1, close=1, cnn=2, even=1, foxnews=1, like=1, network=1, news=1, peoples=1, rating=2
☒ Visualize numeric values		realDonaldTrump	Tonight I will b...	realDonaldTrump	4.61414e-06	0.999995	4.61414e-06	10=1, 10pm=2, andersoncoopers=1, cnn=1, enjoy=1, foxnews=1, sean Hannity=1, tonight=1, trump2016=1, votetrumpsc=1, w=1
☒ Color by instance classes		realDonaldTrump	On my way to L...	realDonaldTrump	4.71826e-06	0.999995	4.71826e-06	iowa=1, makeamericagreatagain=1, nationals=1, news=1, numbers=1, poll=1, received=1, thanks=1, trump2016=1, ways=1
		realDonaldTrump	The dishonest ...	realDonaldTrump	4.94415e-06	0.999995	4.94415e-06	angry=1, bernie=1, crooked=1, deal=1, dishonest=1, looking=1, makes=1, medium=1, mentions=1, sanders=1, speech=1, wa=1, wish=1
		realDonaldTrump	Great new poll...	realDonaldTrump	6.5373e-06	0.999993	6.5373e-06	amp=2, carolina=1, great=1, join=1, let=1, maga=1, movements=1, nevada=1, news=1, north=1, ohio=1, poll=1, thanks=1, today=1
		realDonaldTrump	The media cov...	realDonaldTrump	6.57621e-06	0.999993	6.57621e-06	average=1, clinton=1, cnn=1, conventions=1, covers=1, joke=1, little=1, medium=1, mornings=1, morning=1, sade=1, speech=1, wat=1
		realDonaldTrump	Thank you to ...	realDonaldTrump	6.6492e-06	0.999993	6.6492e-06	also=1, arizona=1, foxandfriends=1, great=2, immigration=1, last=1, night=1, people=1, review=1, speech=1, thank=2
		realDonaldTrump	The rally in Cin...	realDonaldTrump	7.35773e-06	0.999993	7.35773e-06	cancelled=1, cincinnati=1, false=1, makeamericagreatagain=1, medium=1, put=1, rally=1, report=1, trump2016=1, wa=1
		realDonaldTrump	Thank you to NL...	realDonaldTrump	8.48017e-06	0.999992	8.48017e-06	amp=1, another=1, evening=1, great=2, heading=1, last=1, louisiana=1, mi=1, nc=1, reception=1, speech=2, success=1, thank=1, tonight=1
		realDonaldTrump	That was an am...	realDonaldTrump	8.62064e-06	0.999991	8.62064e-06	amazing=1, dishonest=1, foxandfriends=1, hope=1, interview=1, medium=1, nytimes=1, picks=1, rest=1, show=1, totally=1, wa=1
		realDonaldTrump	Lying Cruz put...	realDonaldTrump	9.04188e-06	0.999991	9.04188e-06	3=1, amp=1, crazy=1, cruz=2, dishonest=1, gay=1, liar=1, lying=1, manages=1, obama=1, perhaps=1, put=1, Rubio=1, statements=1, trump=1
		realDonaldTrump	Crooked Hillary...	realDonaldTrump	9.04796e-06	0.999991	9.04796e-06	clinton=1, crooked=1, dishonest=1, expose=1, hillary=1, medium=2, person=1, politician=1, protected=1, refuse=1, talented=1
		realDonaldTrump	#FolNews Poll ...	realDonaldTrump	9.95667e-06	0.99999	9.95667e-06	foxnews=1, makeamericagreatagain=1, poll=1, thanks=1, trump2016=1
		realDonaldTrump	WOW, great ne...	realDonaldTrump	1.03863e-05	0.99999	1.03863e-05	great=1, news=2, newyorkvalues=1, poll=1, support=1, thank=1, trump2016=1, wowe=1, york=1
		realDonaldTrump	*@lopez05391: ...	realDonaldTrump	1.05352e-05	0.999989	1.05352e-05	big=1, crowd=1, jlopez05391=1, love=1, makeamericagreatagain=1, realdonaldrumps=1, rochester=1, see=1, tomorrow=1, trump2016=1, t...
		realDonaldTrump	Great new poll...	realDonaldTrump	1.13564e-05	0.999989	1.13564e-05	great=1, makeamericagreatagain=1, news=1, poll=1, thanks=1, trump2016=1
		realDonaldTrump	Great poll out o...	realDonaldTrump	1.317e-05	0.999987	1.317e-05	americafirst=1, great=1, maga=1, nevada=1, poll=1, see=1, soon=1, thanks=1
		realDonaldTrump	Thank you -Tu...	realDonaldTrump	1.39393e-05	0.999986	1.39393e-05	000=1, 6=1, afternoon=1, arizona=1, great=1, makeamericagreatagain=1, supporter=1, thanks=1, tuccos=1, tuesday=1, votetrump=1
		realDonaldTrump	I am on @FoxN...	realDonaldTrump	1.63836e-05	0.999984	1.63836e-05	enjoy=1, foxnews=1, great=1, halle=1, makeamericagreatagain=1, town=1, trump2016=1, wisconsin=1
		realDonaldTrump	Great POLL nu...	realDonaldTrump	1.72342e-05	0.999983	1.72342e-05	another=1, coming=1, crooked=1, even=1, four=1, great=1, hillary=1, maga=1, numbers=1, obama=1, people=1, poll=1, want=1, worse=1...
		realDonaldTrump	THANK YOU A...	realDonaldTrump	1.79453e-05	0.999982	1.79453e-05	arizona=1, aspirinary=1, gets=1, makeamericagreatagain=1, thank=1, trump2016=1, tuesday=1, votetrump=1
		realDonaldTrump	Great being in...	realDonaldTrump	1.83071e-05	0.999982	1.83071e-05	americafirst=1, ca=1, cincinnati=1, great=1, last=1, night=1, ohio=1, thanks=1, trump2016=1, washington=1
		realDonaldTrump	Great poli: Tha...	realDonaldTrump	1.84437e-05	0.999982	1.84437e-05	10=1, 15=1, 18=2, 3=1, 36=1, 7=1, carolina=1, carson=1, cruz=1, great=1, kasich=1, north=1, poll=1, survey=1, survey=1, trump=1, trum...
		realDonaldTrump	Great new poll L...	realDonaldTrump	1.93682e-05	0.999981	1.93682e-05	great=1, imwithyou=1, iowa=1, makeamericagreatagain=1, news=1, poll=1, thanks=1
		realDonaldTrump	Thank you illin...	realDonaldTrump	1.95499e-05	0.99998	1.95499e-05	13=1, 14=1, 15=2, 28=1, 3=1, 6=1, 8=1, bush=1, carson=1, cruz=1, great=1, illinois=1, kasich=1, news=1, poll=1, Rubio=1, simon=1, siu=1, ...
		realDonaldTrump	Great new num...	realDonaldTrump	1.99816e-05	0.99998	1.99816e-05	great=1, makeamericagreatagain=1, news=1, numbers=1, thank=1, trump2016=1
		realDonaldTrump	New CHV/VIM...	realDonaldTrump	2.0548e-05	0.999979	2.0548e-05	cnn=1, films=1, hampshires=1, news=2, poll=1, released=1, thanks=1, trump2016=1, wum=1
		realDonaldTrump	Such a great ho...	realDonaldTrump	2.02021e-05	0.999979	2.02021e-05	americafirst=1, debate=1, final=1, great=1, honor=1, imwithyou=1, maga=1, movement=1, poll=1, win=1
		realDonaldTrump	Thank you Texa...	realDonaldTrump	2.16969e-05	0.999978	2.16969e-05	000=1, 10=1, amazing=1, makeamericagreatagain=1, supporter=1, texas=1, thank=1, trump2016=1
		realDonaldTrump	Just received fr...	realDonaldTrump	2.20841e-05	0.999978	2.20841e-05	makeamericagreatagain=1, ohio=1, petes=1, peters=1, received=1, thanks=1, trump2016=1, tuesday=1, votetrump=1
		realDonaldTrump	*@JoeNBC: Lat...	realDonaldTrump	2.38825e-05	0.999976	2.38825e-05	1=3, 10=2, 13=1, 14=1, 2=2, 35=1, cruz=1, gops=1, joenbc=1, kasich=1, latest=1, nh=1, poll=1, Rubio=1, tracking=1, trump=1, umas...
		realDonaldTrump	What a great ev...	realDonaldTrump	2.43434e-05	0.999976	2.43434e-05	beats=1, crooked=1, dysfunctional=1, evenings=1, great=1, hillary=1, interesting=1, rigged=1, sanders=1, system=1, totally=1
		realDonaldTrump	Just arrived in ...	realDonaldTrump	2.56167e-05	0.999974	2.56167e-05	arrived=1, hampshires=1, makeamericagreatagain=1, news=1, supporter=1, thank=1, trump2016=1
		realDonaldTrump	Amazing that w...	realDonaldTrump	2.56972e-05	0.999974	2.56972e-05	amazing=1, big=1, leads=1, nbcs=1, news=1, numbers=1, phony=1, poll=2, press=1, q=1, reports=1, today=1, usa=1, wants=1, wjw=1
		realDonaldTrump	Great rally in Fr...	realDonaldTrump	2.65616e-05	0.999974	2.65616e-05	california=1, crowd=1, fresno=1, great=2, rally=1, thank=1, trump2016=1
		realDonaldTrump	Failed President...	realDonaldTrump	2.65045e-05	0.999973	2.65045e-05	1=1, 2=1, amp=1, campaigning=1, candidate=1, cruz=1, endorsing=1, failed=1, john=1, kasich=1, marco=1, mitt=1, presidential=1, romn...
		realDonaldTrump	I will be on @Se...	realDonaldTrump	2.91001e-05	0.999971	2.91001e-05	10pm=1, enjoy=1, foxnews=1, melaniatrumps=1, sean Hannity=1, tonight=1, trump2016=1, w=1, vipirmary=1, wisconsin=1
		realDonaldTrump	*@Johnkirtley: ...	realDonaldTrump	2.955e-05	0.99997	2.955e-05	discourse=1, great=1, healed=1, jobs=1, johnkirtley=1, makeamericagreatagain=1, megynkelly=1, realdonaldrumps=1, thanks=1, wound=1
		realDonaldTrump	Thank you Gov...	realDonaldTrump	3.01387e-05	0.99997	3.01387e-05	amp=1, chairman=1, gops=1, governors=1, imwithyou=1, makeamericagreatagain=1, priebus=1, reince=1, scottwalkers=1, thanks=1
		realDonaldTrump	Great new poll...	realDonaldTrump	3.05442e-05	0.999969	3.05442e-05	america=1, great=1, imwithyou=1, news=1, poll=1, thanks=1, trump2016=1
		realDonaldTrump	Great new poll...	realDonaldTrump	3.16014e-05	0.999968	3.16014e-05	great=1, makeamericagreatagain=1, news=1, poll=1, Texas=1, votetrump=1
		realDonaldTrump	*@holozee: Gre...	realDonaldTrump	3.20015e-05	0.999968	3.20015e-05	cnn=1, danscavinos=1, slozee=1, areats=1, interview=1, makeamericagreatagain=1, newdavis=1, oaths=1, paulmanfort=1, realdonaldrump...